

# Ian Nelson

## ENGINEERING PORTFOLIO

ANTICIPATED MAY 2028:

Bachelor's Degree in Mechanical Engineering at Liberty University

509.212.7650 | [ianinafrica9@gmail.com](mailto:ianinafrica9@gmail.com) | [www.linkedin.com/in/ian-nelson-95b767352](https://www.linkedin.com/in/ian-nelson-95b767352)

## FORMULA SAE STEERING AND SUSPENSION ENGINEER

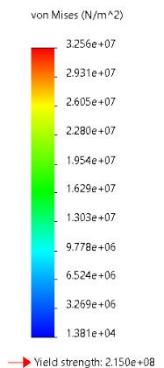
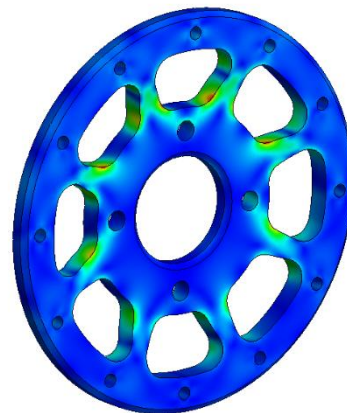
FALL 2024 – CURRENT

### SKILLS:

- CAD Modeling (SolidWorks)
- Design and Analysis of Suspension Systems
- Machining

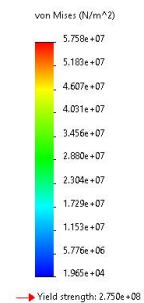
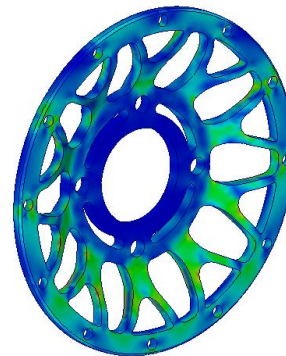
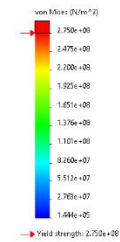
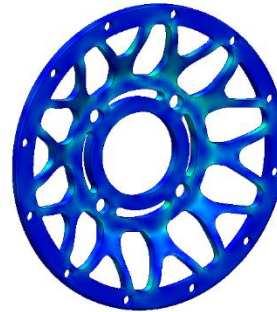
### FSAE WHEEL CENTER DESIGN

YEAR 1 DESIGN (2024-2025 CAR)



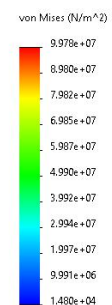
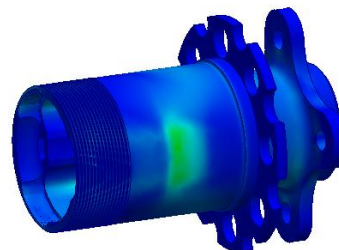
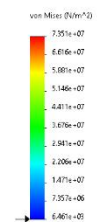
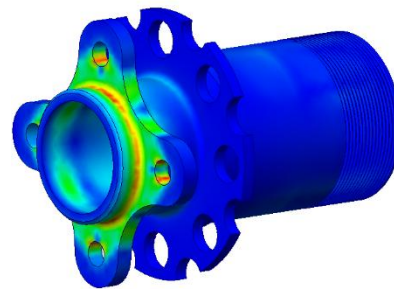
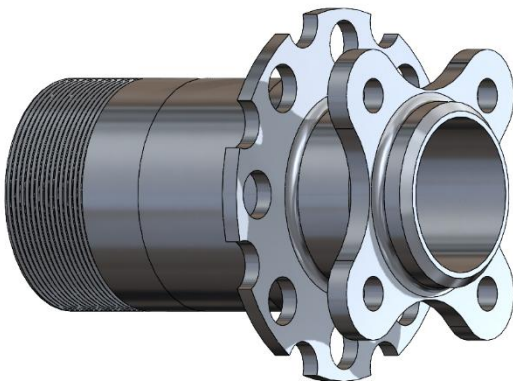
## YEAR 2 DESIGN (2025-2026 CAR)

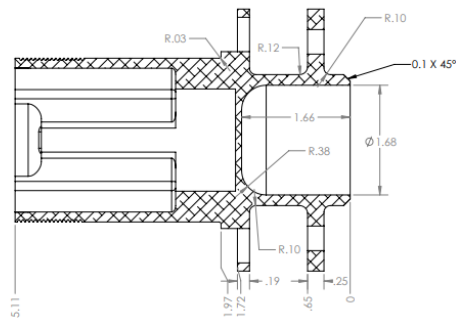
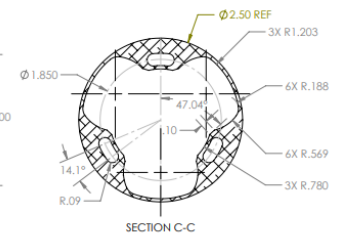
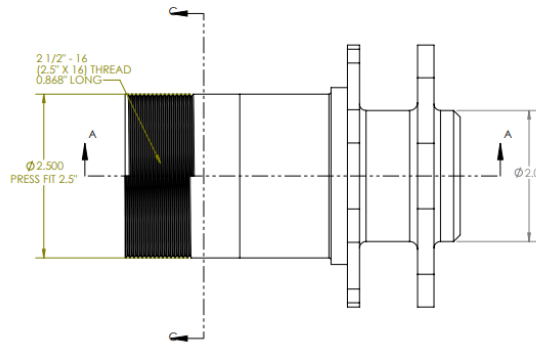
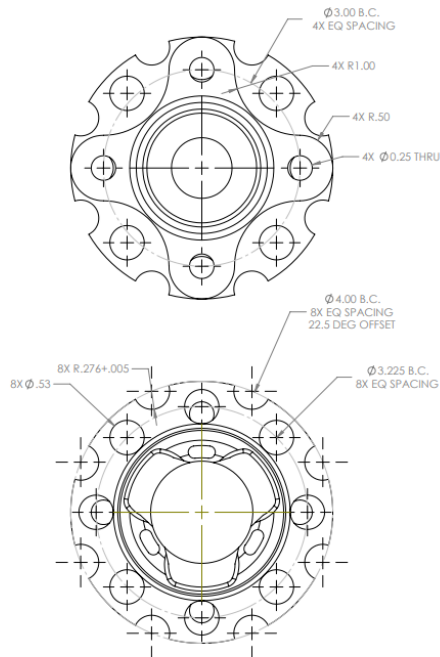
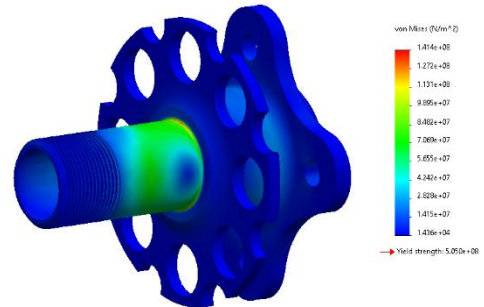
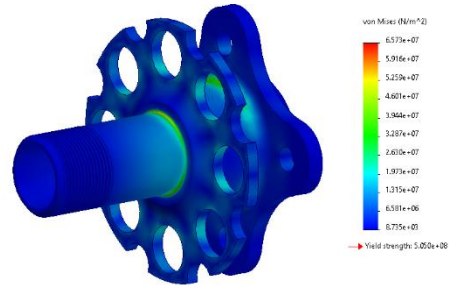
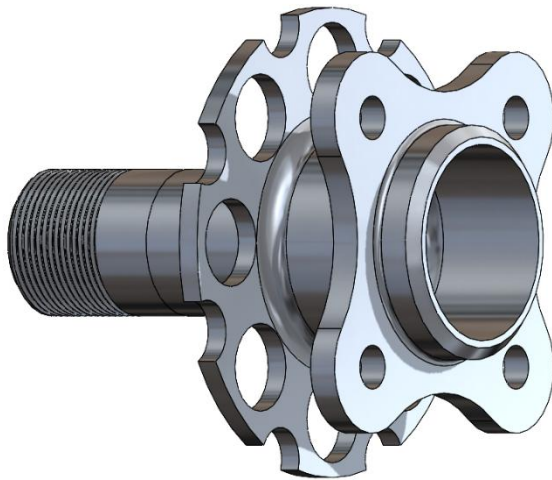
- Weight reduction of 50% from the previous year.
- Refined FEA simulations by using on-car data acquisition to derive maximum lateral and longitudinal G-loading.



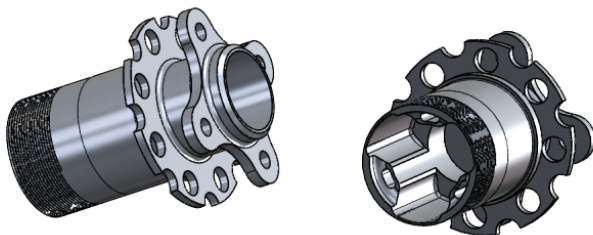
## FSAE WHEEL HUB DESIGN

- 30-40% weight reduction between front and rear hubs.
- Iterated design to balance weight and simulated strength.





SECTION A-A



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|  |             |                                                                                                                                                                  |
|--|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |             | UNLESS OTHERWISE SPECIFIED:                                                                                                                                      |
|  |             | DIMENSIONS ARE IN INCHES<br>TOLERANCES:<br>FRACTIONAL: $\frac{1}{16}$<br>ANGULAR: MACHINE $\pm 10^\circ$<br>TWO PLACE DECIMAL<br>THREE PLACE DECIMAL: $\pm .001$ |
|  |             | INTERPRET GEOMETRIC<br>TOLERANCING PER:<br>NATURAL<br>ALUMINUM 7075-T6                                                                                           |
|  | NEXT ASSY   | USED ON: <i>Handwritten: 100000</i>                                                                                                                              |
|  | APPLICATION | DO NOT SCALE DRAWING                                                                                                                                             |

| TITLE: FLAMES RACING REAR HUB |          |         |              |
|-------------------------------|----------|---------|--------------|
| SIZE                          | DWG. NO. | REV     |              |
| C                             | 1        |         |              |
| SCALE: 1:1                    |          | WEIGHT: | SHEET 1 OF 1 |

## FSAE WHEEL HUB FINAL PRODUCT:

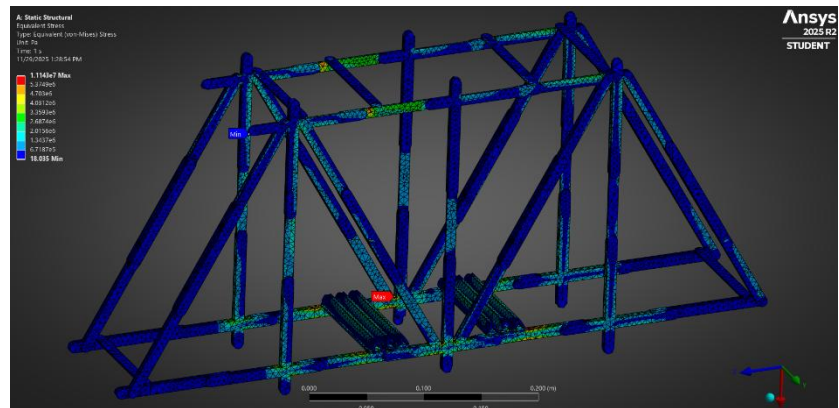
- Correctly Machined Hub  
(Left): Final component manufactured to exact specifications.
- Incorrectly Machined Hub  
(Right): Hub sheared during use. Cause: A chamfer was incorrectly machined in the place of a fillet, creating a stress concentration which caused the part to fail.



## SCHOOL PROJECTS

### STATICS POPSICLE STICK BRIDGE

- Predicted bridge load capacity within ~5% based on static structural Ansys simulation.



## MECHANICAL ENGINEERING INTERNSHIP – VALMET INC.

- Created SolidWorks models for tube shield bend die designs.
- Introduced an Excel-based sizing tool to output bend die dimensions for irregular tube shield sizes accounting for material spring-back.